

Windows Server & Active Directory

Complete Study Notes

Including: API & Application Programming Interface

Network Operating System • Active Directory • GPO • API

Subject: Windows Server
Administration

Level: ICT / Networking

Topics: 11 Core Sections

1 Network Operating System (NOS)

A **Network Operating System (NOS)** is an operating system specifically designed to manage resources and services across a network, allowing multiple computers and users to share files, printers, and internet connections from a central location.

Key Differences from Regular OS:

Feature	Regular OS (Windows 11)	Network OS (Windows Server)
Users served	1 user at a time	Hundreds/thousands of users
Purpose	Personal use	Centralised network management
Security	Local user accounts	Domain-wide policies & AD
Resources	Local only	Shared across the network

Common Examples:

- Windows Server 2019 / 2022 / 2025 — Microsoft's enterprise NOS
- Ubuntu Server / CentOS — Linux-based NOS
- Novell NetWare — older but historically important

2 Requirements for Server Operating Systems

Hardware Requirements

Component	Minimum	Recommended	Notes
CPU	1.4 GHz 64-bit	3.1 GHz+ multi-core	64-bit required
RAM	512 MB	16 GB+	More = better performance
Storage	32 GB	100 GB+ SSD	SSD preferred
Network	Basic NIC	Gigabit NIC	Redundant NICs ideal
Display	800×600 (remote)	1920×1080	Often headless

Software Requirements

- 64-bit processor architecture is mandatory for modern Windows Server
- Compatible and updated device drivers for all hardware components
- Valid Windows Server license (Standard or Datacenter edition)
- Latest BIOS/UEFI firmware updates before installation
- Client PCs must run Windows 10/11 **Pro or Enterprise** (NOT Home — Home cannot join a domain)
- DNS must be configured to point to the Domain Controller

3 Active Directory (AD)

Active Directory (AD) is Microsoft's directory service that stores and manages information about all users, computers, printers, and resources on a network. It acts as a central database that controls who can log in, what they can access, and what rules apply to them. It runs on a Domain Controller and uses protocols such as LDAP, Kerberos, and DNS.

What Active Directory Does:

- **Authentication** — Verifies identity when users log in (Kerberos tickets)
- **Authorisation** — Controls what resources each user/computer can access
- **Centralised Management** — Manage all accounts from one place
- **Policy Enforcement** — Apply security rules via GPO to all machines
- **DNS Integration** — AD depends on DNS to locate domain controllers

***Simple Analogy:** Think of AD as the HR department + security office of a company — it knows every employee, controls their access badge, and enforces company rules.*

4 Active Directory Objects

Everything stored inside Active Directory is called an **Object**. Each object represents something real on the network and has **attributes** (properties like name, email, IP address, etc.).

Object Type	Represents	Key Attributes
User	A person's account	Username, password, email, department
Computer	A machine on the domain	Computer name, OS version, IP
Group	Collection of users/computers	Members, scope, type
OU	Organisational folder in AD	Name, linked GPOs, delegated admin
Printer	Shared network printer	Location, driver, share name
Contact	External person (no login)	Name, email, phone

5 Forest

A **Forest** is the highest-level boundary of Active Directory. It is a collection of one or more domains that share a common **schema** (the same rules for object structure) and a common **Global Catalog**. All domains in a forest automatically trust each other.

Forest Structure Example:

```
FOREST: masu.local
|
+-- DOMAIN: masu.local (Root Domain)
```

```
|  
+-- DOMAIN: hr.masu.local (Child Domain)  
|  
+-- DOMAIN: it.masu.local (Child Domain)
```

Key Points:

- One forest = one schema (shared object structure rules)
 - Domains within a forest have **automatic two-way trust**
 - Most companies use a **single forest**
 - Multiple forests = no automatic trust (used when companies merge)
 - The first domain created becomes the **Forest Root Domain**
-

6 Domain Controller (DC)

The **Domain Controller (DC)** is the server that runs Active Directory Domain Services (AD DS). It is the brain of the domain — it stores the AD database, authenticates users at login, enforces Group Policies, and handles DNS for the domain.

What the Domain Controller Does:

- Stores the AD database file: **NTDS.dit**
- Authenticates user logins using the **Kerberos** protocol
- Replicates AD data to other DCs for fault tolerance
- Enforces Group Policy Objects (GPOs) to users and computers
- Runs DNS to resolve domain names within the network

Best Practice: Always have at least 2 Domain Controllers. If the only DC goes offline, no one on the network can log in.

FSMO Roles (Special DC Responsibilities):

FSMO Role	Responsibility
PDC Emulator	Handles password changes, time sync, legacy auth
RID Master	Assigns unique ID blocks to other DCs
Infrastructure Master	Updates references between domains
Schema Master	Controls changes to the AD schema (forest-wide)
Domain Naming Master	Manages adding/removing domains (forest-wide)

7 Organisational Unit (OU), Group & User

Organisational Unit (OU)

An **OU** is a container (folder) inside Active Directory used to organise objects logically — usually by department, location, or function. OUs are the main targets for applying **Group Policy Objects** and for **delegating administrative control**.

```
masu.local
|
+-- OU: IT Department
|   +-- User: Maty
|   +-- Computer: IT-PC-01
|   +-- Group: IT_Staff
|
+-- OU: Finance
|   +-- User: Sara
|   +-- Group: Finance_Team
```

A screenshot of the Active Directory console. The left pane shows a tree view with 'OU: HR' selected. The right pane is empty.

|
+-- OU: HR

Group

A **Group** is a collection of user accounts, computer accounts, or other groups. Instead of assigning permissions to 50 individual users, you assign permissions once to the group — all members automatically inherit those permissions.

User

A **User** object is an individual login account in Active Directory. It stores credentials (username + password) and personal information (name, email, department, phone). Users are placed inside OUs and added to Groups.

8 Scope — Security Group & Distribution Group

Group Scope

Scope defines how wide a group's membership and permissions reach across the domain or forest.

Scope	Members Can Come From	Assign Permissions In	Best Used For
Domain Local	Any domain in forest	Same domain only	Assigning access to local resources
Global	Same domain only	Any domain in forest	Grouping users with similar roles
Universal	Any domain in forest	Any domain in forest	Multi-domain environments

Security Group vs Distribution Group

Type	Purpose	Can Assign Permissions?	Use Case
Security Group	Access control	YES	Folder access, printer access, login rights
Distribution Group	Email mailing list	NO	Send one email to all members of a team

Remember: Security Groups are the ones you use for everything access-related. Distribution Groups are only for email — they cannot control resource access.

9 Account Policy & Joining a Client to the Domain

Password Policy

Setting	Recommended Value	What It Means
Minimum Password Length	8+ characters	Password must be at least 8 characters
Password Complexity	Enabled	Must include uppercase, number, symbol
Maximum Password Age	90 days	User must change password every 90 days
Minimum Password Age	1 day	Cannot change password again immediately
Password History	10 passwords	Cannot reuse the last 10 passwords

Account Lockout Policy

Setting	Recommended Value	What It Means
Lockout Threshold	5 attempts	Account locks after 5 wrong passwords
Lockout Duration	30 minutes	Account stays locked for 30 minutes
Reset Counter After	15 minutes	Failed attempt count resets after 15 min

Requirements to Join a Client Computer to the Domain

■ OS must be **Windows Pro or Enterprise** — Home edition cannot join a domain

- The client computer must be on the **same network** as the Domain Controller
- **DNS** on the client must point to the Domain Controller's IP address
- You need **Domain Admin credentials** to authorise the join
- The computer name must be **unique** on the network

Steps to Join a Domain (Windows 10/11):

1. Settings > System > About
2. Click 'Rename this PC (Advanced)'
3. Under 'Member of' select 'Domain'
4. Type the domain name e.g. masu.local
5. Enter Domain Admin username and password
6. Restart the computer

10 Server Roles

Roles are services you install on Windows Server to give it specific jobs. A server can hold multiple roles. Roles are installed via **Server Manager > Add Roles and Features**.

Role Name	What It Does	Common Use
AD DS (Active Directory Domain Services)	Makes the server a Domain Controller	Required — core of AD
DNS Server	Resolves domain names to IP addresses	Required with AD DS
DHCP Server	Automatically assigns IP addresses to clients	Replaces manual IP config
File & Storage Services	Shares folders and drives on the network	Network file sharing
Print Services	Manages shared network printers	Centralised printing
Web Server (IIS)	Hosts internal or external websites	Intranet portals
Remote Desktop Svc	Lets users connect to server remotely	Remote work
Hyper-V	Runs virtual machines on the server	Server virtualisation

11 GPO — Group Policy Object

A **Group Policy Object (GPO)** is a set of rules and settings that are automatically applied to users and computers in Active Directory. GPOs let administrators enforce security and configuration settings across the entire organisation from one central place.

What GPOs Can Control:

- Password and account lockout policies
- Desktop wallpaper, screensaver, and Start menu settings
- Block USB drives from being used on company computers
- Automatically map network drives when a user logs in
- Install or block specific software applications
- Restrict access to Control Panel or Settings
- Configure Windows Firewall rules
- Set browser homepage and proxy settings

GPO Application Order — LSDOU (most specific wins):

Order	Level	Scope
1 (first applied)	Local Policy	Settings on the individual PC itself
2	Site Policy	Applies to an AD Site (physical location)
3	Domain Policy	Applies to the entire domain
4 (last applied = wins)	OU Policy	Applies to objects inside a specific OU

Rule: If two GPOs conflict, the one applied LAST wins. OU Policy overrides Domain Policy.

Useful GPO Commands:

```
gpupdate /force -- Force-apply all GPOs immediately
gpresult /r -- Show which GPOs are applied to current user/PC
gpmc.msc -- Open Group Policy Management Console
rsop.msc -- Resultant Set of Policy (shows effective settings)
```

12 API — Application Programming Interface

What is an API?

An **API (Application Programming Interface)** is a set of rules and tools that allows one software application to communicate with another. Think of it as a **waiter in a restaurant**: you (the application) tell the waiter (the API) what you want, the waiter takes the request to the kitchen (the system/server), and brings back the result — without you needing to know how the kitchen works.

Why APIs Matter in Windows Server:

In the context of Windows Server and Active Directory, APIs allow administrators and developers to **automate tasks, manage users/resources programmatically, and integrate AD with other software** — without manually clicking through the Server Manager GUI every time.

Types of APIs in Windows Server Context:

API Type	Full Name	What It Does
Win32 API	Windows 32-bit API	Core Windows OS functions — file access, processes, networking
ADSI	Active Directory Service Interfaces	Lets apps read/write AD objects (users, groups, OUs)
LDAP API	Lightweight Directory Access Protocol API	Query and modify AD over the network
WMI	Windows Management Instrumentation	Manage and monitor Windows hardware/software via scripts
REST API	Representational State Transfer API	Web-based API used by modern cloud management tools
Graph API	Microsoft Graph API	Manage Azure AD / Microsoft 365 via HTTP requests

ADSI — Active Directory Service Interfaces

ADSI is the most important API for Windows Server students. It is a set of COM interfaces built into Windows that allows programs and scripts to interact directly with Active Directory — creating users, resetting passwords, querying group membership, and more.

Example — Creating a User with PowerShell (uses ADSI under the hood):

```
# PowerShell uses AD APIs to create a new user
New-ADUser -Name 'Maty Haile'
-SamAccountName 'maty.haile'
-UserPrincipalName 'maty.haile@masu.local'
-AccountPassword (ConvertTo-SecureString 'P@ss123' -AsPlainText -Force)
-Enabled $true
```

REST API — How Modern Server Management Works

A **REST API** uses standard **HTTP methods** to send requests to a server and get responses (usually in JSON format). Modern Windows Server management tools, Azure, and Microsoft 365 all expose REST APIs.

HTTP Method	Action	Example Use
GET	Read / Retrieve data	Get a list of all users in AD
POST	Create new data	Create a new user account
PUT	Update existing data	Change a user's department
DELETE	Remove data	Delete a computer account from AD

Microsoft Graph API

The **Microsoft Graph API** is Microsoft's unified REST API for managing Azure Active Directory (now called Entra ID), Microsoft 365, Teams, OneDrive, and other Microsoft cloud services. It uses HTTPS and returns JSON data. Endpoint base URL: <https://graph.microsoft.com/v1.0/>

Example Graph API Requests:

```
GET https://graph.microsoft.com/v1.0/users
-- Returns a list of all users in the organisation

GET https://graph.microsoft.com/v1.0/users/maty@masu.local
-- Returns details of a specific user

POST https://graph.microsoft.com/v1.0/groups
-- Creates a new group (body contains group details in JSON)
```

WMI — Windows Management Instrumentation

WMI is a Windows API that provides a standardised way to access system information and manage Windows components — hardware, software, OS settings, running processes, and network configuration. It is heavily used in enterprise monitoring and automation scripts.

WMI Examples (PowerShell):

```
# Get OS information via WMI API
Get-WmiObject -Class Win32_OperatingSystem

# List all running processes
Get-WmiObject -Class Win32_Process

# Get network adapter info
Get-WmiObject -Class Win32_NetworkAdapterConfiguration
```

How APIs Connect to Everything Else:

```
Administrator / Script / App
|
| (API Call)
v
```

```
Windows API / ADSI / REST / Graph API
|
| (Processes Request)
v
Active Directory / Windows Server / Azure
|
| (Returns Result)
v
Application receives data / confirmation
```

API Summary Table:

API	Where Used	Protocol	Key Use in Server Admin
Win32 API	Local Windows apps	OS calls	System-level programming
ADSI	AD management scripts	COM/LDAP	Create/manage AD objects
LDAP	Any AD-aware app	TCP port 389	Query AD directory data
WMI	Windows automation	WMI/DCOM	Hardware & OS monitoring
REST API	Web & cloud apps	HTTPS/JSON	Modern server management
Graph API	Azure / M365 mgmt	HTTPS/JSON	Cloud AD management



Quick Reference Cheat Sheet

Key Terms at a Glance

Term	One-Line Definition
NOS	OS that manages shared resources across a network
Active Directory	Central database of all network users, computers, and resources
Domain Controller	Server that runs AD — authenticates users and enforces policy
Forest	Top-level AD boundary containing one or more domains
Domain	Logical group of network objects sharing an AD database
OU	Folder in AD used to organise objects and apply GPOs
User	Individual login account in AD
Group	Collection of users/computers for easier permission management
Security Group	Group used to assign resource access permissions
Distribution Group	Group used only as an email mailing list
GPO	Set of rules automatically applied to users/computers in AD
API	Interface that lets software applications talk to each other
ADSI	Microsoft API for accessing and managing Active Directory
Graph API	Microsoft's REST API for managing Azure AD and Microsoft 365
WMI	Windows API for monitoring and managing system components
LDAP	Protocol used to query and update directory services like AD
FSMO	Special roles held by Domain Controllers for specific AD tasks
LSDOU	GPO application order: Local, Site, Domain, OU

Essential Commands

```
gpupdate /force Force-apply all Group Policies immediately
gpresult /r Show GPOs currently applied to the user/computer
gpmc.msc Open Group Policy Management Console
dsa.msc Open Active Directory Users and Computers
dcpromo Promote a server to Domain Controller (older method)
nltest /dclist: List all Domain Controllers in a domain
nslookup Test DNS resolution (critical for domain joining)
net user Manage local user accounts from command line
ipconfig /all Show IP config including DNS server (must point to DC)
```

These notes cover all 12 topics for your Windows Server exam. The most important practical skills to practise are: (1) joining a client PC to the domain, (2) creating users and groups in AD, (3) applying a GPO to an OU, and (4) understanding what an API is and how ADSI/Graph API interact with AD.